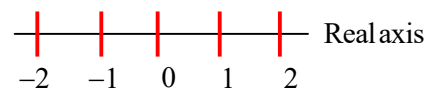


Complex Analysis

5.1 Origin of a complex number

$$\sqrt{4} = +2, -2$$

$$\sqrt{-4} = \sqrt{4}\sqrt{-1} = \sqrt{4}i = 2i, -2i$$



$$z = x + iy \Rightarrow z = (x, y) \text{ or } (r, \theta)$$

$$z = r \cos \theta + ir \sin \theta$$

$$\Rightarrow r = \sqrt{x^2 + y^2} \text{ (Modulus)} \Rightarrow \theta = \tan^{-1} \left(\frac{y}{x} \right) \text{ (Argument)}$$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$